Triglycerides

Triglycerides are composed of glycerol with fatty acids esterified to all the 'C's. They do lack polar head group but they are not present in membrane.

Lipoproteins are denoted as proteins that are chemically attached to membrane lipids, and they are present on either sides of a membrane.

Water, the universal solvent is also a component of a membrane. The cell membranes include water molecules, either bound to the polar groups visible on the polar side of the membrane molecules (structured water) or unbound (bulk water) within pores and certain ion channels crossing the membranes .

Ions are associated with membranes by simple adsorption to the two surfaces of the membrane or can just pass through ionic channels (membrane proteins) or ion pumps (membrane proteins with enzymatic character). The ions that contributes to the structure and functions of membrane are: H^+, Na^+, K^+, Cl^-, Ca^{++}, HCO_3^-

1.1.2.Proteins

Membrane proteins

Most of the functions of a membrane depend on the proteins present in it. You can recall that it was mentioned elsewhere that there are two classes of membrane proteins.

Both integral and peripheral membrane proteins which form the structure of the membrane. Integral proteins have both hydrophobic and hydrophilic interaction with hydrocarbon core and exterior of lipid cage

1.1.2.1.Integral Protein

Some of the membrane proteins are tightly embedded in the membrane and they cannot be isolated unless, the membrane is disintegrated. They are called as Integral or Intrinsic membrane proteins. They are again classified into two. (a). Transmembrane proteins, which traverse (pass through) or span the membrane. These proteins will have domains on either side of the membrane. Many cell surface receptors belong to this class. (b). Lipid anchored proteins that are present either on the cytosolic side or on the extracytosolic side. They insert themselves in the membrane by a lipid (acyl chain) attached to the N terminal end. Transmembrane proteins are of two types. Single pass transmembrane proteins that traverse the membrane only once. Multipasstransmembrane proteins that traverse the membrane more than once.

1.1.2.2.Peripheral Protein

Those proteins that are present on the surface of the membrane are called as peripheral proteins. They can be easily isolated from the membrane. eg. spectrin present in the RBC membrane.

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